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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations Oct-2020

Electronics and Communication Engineering

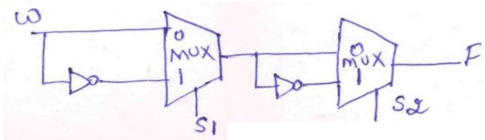
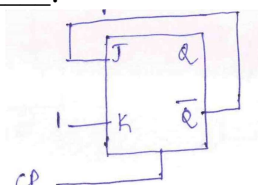
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

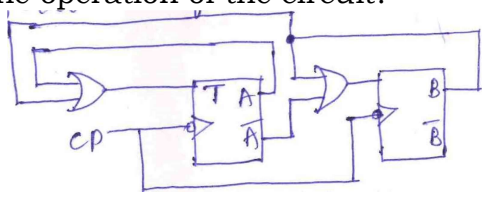
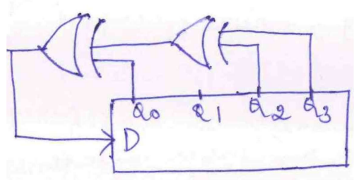
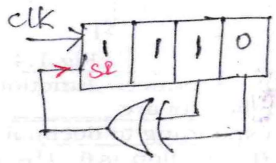
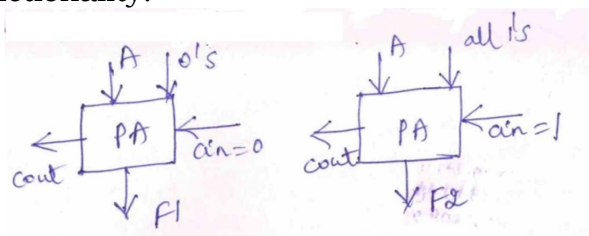
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

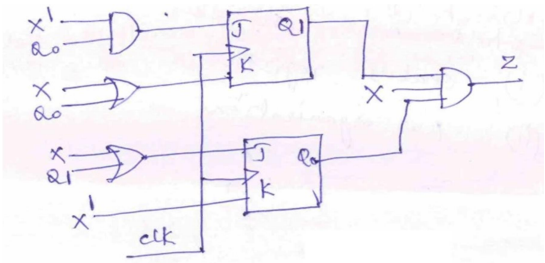
PART-A

1	1.1	A $74LS00$ $NAND$ gate output is driving three $74LS00$ and two $74AS00$ $NAND$ gates. Determine whether there is a loading problem. Different gate specifications are shown below: $74LS00$ $74AS00$ IOH $-0.4mA$ $-2mA$ IIH $20\mu A$ $20\mu A$ IOL $8mA$ $20mA$ IIL $-0.4mA$ $-0.5mA$	02
	1.2	If the output high current of a gate is $100mA$ and the input high current of the driven gate$10mA$, then the fan out of the driving gate is _____.	01
	1.3	In a 5 variable K-map, cell 3 is adjacent to _____, _____, _____, _____ and _____.	01
	1.4	The functionality (F) implemented using combinational circuit shown in fig 1.4 is _____.  Fig 1.4	02
	1.5	In the sum of product function, $f(A,B,C) = \sum_m(2,3,4,5,7)$, the essential prime implicants are _____.	01
	1.6	The maxterm corresponding to decimal number 15 is _____.	01
	1.7	The Q output of SR flip flop is 0. The output does not change when clock pulse is applied. The input S and R will be respectively _____ and _____.	01
	1.8	A JK flip flop is connected as shown in fig 1.8. Assuming the flip flop was initially cleared and then clocked for 6 pulses, the sequence at the 'Q' output will be _____.  	

1.9	A ripple counter consists of 8 FFs each with propagation delay of $10nS$. What is the maximum clock frequency of the counter for reliable operation?	01
1.10	The sequential circuit in fig 1.10 consists of two flip flops with initial state 00. What is the operation of the circuit?  Fig 1.10	02
1.11	If a counter having 11 FFs is initially at 0, the value of the count it holds after 2064 pulses is _____.	01
1.12	A 4 bit right shift register is initialized to value 1000. (Q_3, Q_2, Q_1, Q_0) as shown in fig 1.12. The pattern 1000 will appear after how many clock cycles?  Fig 1.12	02
1.13	The initial content of the 4-bit serial in parallel out right shift register shown in fig 1.13 is 1110. After three clock pulses, the content of the shift register will be _____.  Fig 1.13	02
1.14	Check for the equivalence of the functions F_1 and F_2 in fig 1.14. Identify the functionality.  Fig 1.14	02

PART-B

2	a	Explain the working of $ECL\text{-}NOR$ gate with the help of a neat circuit diagram.	06
	b	Simplify the following Boolean function using k-map and obtain the most optimal SOP expression. $$f(a,b,c,d,e) = \sum_m (0,1,4,5,6,7,9,12,13,14,15,16,17,20,21,22,23,25,29)$$	05
	c	Simplify $F(w,x,y,z) = \sum_m(1,4,6,7,8,9,10,11,15)$ using Quine-McCluskey method.	05
3	a	Design a 4-bit carry lookahead adder. Comment on the circuit propagation delay.	06

	b	Design an application specific decoder to drive common cathode seven segment display.	05
	c	Design and implement a BCD adder circuit using binary adders.	05
		OR	
4	a	Assume that a room has three doors and a switch near by each door to control a single light in the room. Let X, Y and Z denote the state of all the switches. Assume the light is off if all the switches are open. Closing any switch turns the light ON, but closing another switch turns the light OFF. Light is ON if all the switches are closed. Construct a truth table and implement the logic using $2:1$ multiplexer along with minimum possible number of gates.	05
	b	Design a 4 to 2 priority encoder using basic gates, where order of priority levels are given as I_3, I_0, I_1, I_2 for the inputs I_3, I_2, I_1 and I_0.	05
	c	A magnitude comparator has $A(A_3A_2A_1A_0)$ and $B(B_3B_2B_1B_0)$ as inputs. Obtain the Boolean functions for $A > B, A < B$ and $A = B$. Show the realization using basic gates.	06
5	a	Design a 4-bit shift register with a parallel load using D flip-flops. There are two control inputs: shift and load. When shift = 1 and load= 0, the contents of the register are shifted by one position right. New data is transferred to the register when load = 1 and shift = 0. If both control inputs are equal to 0 the contents of the register do not change.	06
	b	Design 4-bit serial adder and explain the operation with suitable example.	05
	c	With a neat diagram and function table explain the working of master slave JK flip flop with preset and clear facilities; use $NAND$ gates only.	05
		OR	
6	a	Design a positive edge triggered D-flip flop using $NAND$ gates and explain its working with relevant waveforms and function table.	05
	b	Design a sequence generator to generate the binary sequence 100010011010111.	05
	c	Obtain the governing equations and the state table for fig 6c. Also write the state diagram.  Fig 6c	06
7	a	Design a synchronous counter to count $000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 101 \rightarrow 110 \rightarrow 111 \rightarrow 000$ using JK flip flops.	06
	b	Design a 3-bit binary counter with parallel load facility. Also explain the working of 3-9 and 4-6 counters using IC 74192.	10

8	a	Design an arithmetic logic unit with three selection variables $S_2S_1S_0$ that generate the following operations: draw the logic diagram of one typical stage.	10

	Arithmetic Operations		*Logical Operations*	
$S_2S_1S_0$	$c_{in} = 0$	$c_{in} = 1$	$S_2S_1S_0$	$c_{in} = X$
000	$F = A$	$F = A + 1$	100	OR
001	$F = A + B$	$F = A + B + 1$	101	XOR
010	$F = A - B - 1$	$F = A - B$	110	AND
011	$F = A - 1$	$F = A$	111	$Complement$

	b	With relevant diagrams explain: i) Status register ii) Bus organization.	06